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Research Paper

Differential engagement of attention and visual working memory in the representation and evaluation of the number of relevant targets and their spatial relations: Evidence from the N2pc and SPCN

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ABSTRACT

We examined the role of attention and visual working memory in the evaluation of the number of target stimuli as well as their relative spatial position using the N2pc and the SPCN. Participants performed two tasks: a simple counting task in which they had to determine if a visual display contained one or two coloured items among grey fillers and one in which they had to identify a specific relation between two coloured items. The same stimuli were used for both tasks. Each task was designed to permit an easier evaluation of either the same-coloured or differently-coloured stimuli. We predicted a greater involvement of attention and visual working memory for more difficult stimulus-task pairings. The results confirmed these predictions and suggest that visuospatial configurations that require more time to evaluate induce a greater (and presumably longer) involvement of attention and visual working memory.

1. Introduction

We are frequently exposed to numerous incoming stimuli that can overload recognition and memory mechanisms (Sperling, 1960). As such, the ability to select the most relevant stimuli to be processed to a higher level is a critical aspect of normal cognitive function. Selective attention and the cognitive functions associated with it have been studied extensively over the years using chronometric methods complemented by measures of brain activity (Pashler, 1988). Event-related potential (ERP) studies have found specific patterns of brain electric activity that are associated with spatial attention and visual working memory, two cognitive functions critical for this type of processing. These components, measured using well-established methods, are now easily reproducible making them an efficient way to study the evolution of sensory and cognition operations over time under different experimental conditions. Here we will focus on two components, the negativity 2 posterior contralateral (N2pc) and the sustained contralatedal negativity (SPCN).

The N2pc (Luck & Hillyard, 1994) is a lateralized component characterized by an enhanced negativity at occipital sites contralateral to the visual hemifield where a relevant item is presented and where attention is presumed to be engaged. The N2pc usually peaks between 200 and 280 ms post-target onset and is usually maximal in amplitude near electrodes PO7 and PO8. It appears to originate in lateral portions of the visual extrastriate cortex (Hopf et al., 2000). There is solid evidence that the N2pc reflects neuronal activity related to attentional selection, although the exact underlying mechanisms are still under debate. Many researchers consider the N2pc as a moment-to-moment index of visuo-spatial attention. Some evidence suggests that the amplitude of N2pc reflects the number of distinct representations that are selected and individuated for further processing (Mazza & Caramazza, 2011). The N2pc has been used to track the deployment of visual spatial attention in a number of situations (e.g., as a reflection of attentional capture, Brisson, Leblanc, & Jolicoeur, 2009; Hickey, McDonald, & Theeuwes, 2006; Kiss & Eimer, 2011; Leblanc, Prime, & Jolicoeur, 2008).

The amplitude of the N2pc has been shown to depend on the nature of the attentional processing required by the task. For example, Mazza and Caramazza (2011) asked subjects to perform three

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different tasks using the same set of stimuli (red and green diamonds on a black background). In one task they reported the number of stimuli presented in a particular colour (e.g., how many red items were in the display?); in another task they reported whether the display contained stimuli in a specific colour, regardless of their number; and in a third task they reported whether a specific number of coloured stimuli are present or not (e.g., were there exactly 2 red items?). Their results showed a modulation of the amplitude of the N2pc for target number (how many items) for the first and third tasks, but not for the second one. The authors took these results to suggest that a coarse initial representation of the relevant objects is created to permit the individuation of each of them by the visual system; a task-dependent process that can be affected by top-down influence. In their view, N2pc reflects the individuation process.

The SPCN (Jolicoeur, Sessa, Dell'Acqua, & Robitaille, 2006) is also a lateralized component seen at occipital sites contralateral to the presentation of encoded stimuli. The SPCN is also called the contralateral negative slow wave (CNSW, Klaver, Talsma, Wijers, Heinze, & Mulder, 1999), or contralateral delay activity (CDA, Vogel & Machizawa, 2004). It is a later component with an onset usually after 300 ms post-target with a maximum also often at the PO7 and PO8 electrodes. The duration (hence offset) of the SPCN has been shown to depend on the duration of processing of information held in visual working memory, either by virtue of the duration of a retention interval (e.g., Perron et al., 2009) or by the duration the processing of information held in visual working memory (e.g., Prime & Jolicoeur, 2010). The SPCN has been linked to active maintenance of information in visual working memory in experiments in which the amplitude of the SPCN increased as the number of items held in memory increased. Importantly, this increase reached a ceiling or plateau when the number of items corresponded to the maximum capacity of the subjects in the experiment (Robitaille et al., 2010; Vogel & Machizawa, 2004). Passage into visual working memory appears to be spontaneously required when a task requires detailed form discrimination for patterns that are not highly overlearned (e.g., Kiss & Eimer, 2011). The SPCN has also been shown to be functionally dissociable from the N2pc, despite very similar scalp distributions of voltage fields (e.g., Jolicoeur, Brisson, & Robitaille, 2008). The SPCN has also been observed in tasks in which the stimuli are continuously visible, as in the curve-tracing task (Lefebvre, Jolicoeur, & Dell'Acqua, 2010). Still, much work remains to be done to understand the exact functional correlates of the N2pc and the SPCN, and of their interrelations.

Experiments on visuospatial attention and working memory are typically based on experiments in which each trial involves the presentation of one visual array used by the subject to base an immediate response. We will call this approach the single-frame procedure. In the present work we used a multiple frame procedure (MFP), in which each trial involved the presentation of six frames followed by the subject's response. The response was based on how many times a specific target was presented in the preceding set of six frames. This procedure takes advantage of the high temporal resolution of electroencephalography (EEG) and the fact that EEG can be used to monitor sensory, perceptual, and cognitive process as they unfold even in the absence of an overt response (see Bentin et al., 2007 for an example). The MFP allows us to multiply the number of events we are able to present to the participants while keeping the length of the study similar to one conducted with a single frame procedure. The activity we are interested in occurs shortly after the presentation of each frame, and the MFP in conjunction with EEG recordings, allowed us to examine attentional, visual working memory, and downstream cognitive operations more efficiently than with single-frame methods (see Sagiv & Bentin, 1999; Vogel & Luck, 2000; for other examples of procedures based on target counting). The MFP was used in previous work, to study

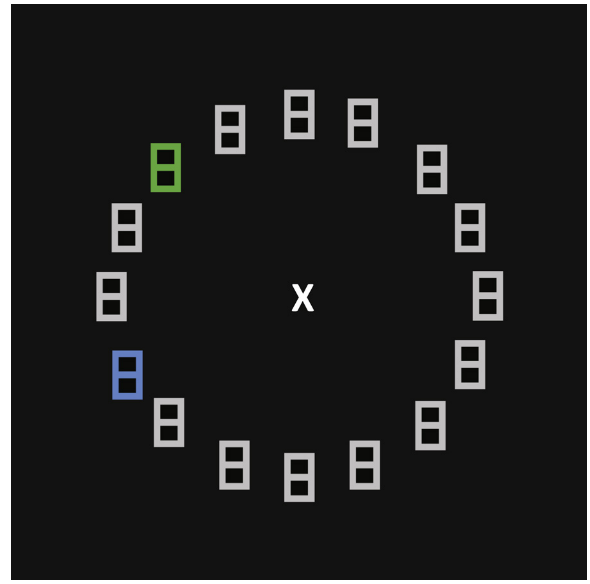

Fig. 1. Example a stimulus display in a frame with two coloured items. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

colour-specific effects on attentional deployment, or conceptual and physical similarity of targets and distractors, with a focus on the N2pc component, providing strong support for the method (Aubin & Jolicoeur, 2016; Drisdelle, Aubin, & Jolicoeur, 2017; Pomerleau, Fortier-Gauthier, Corriveau, Del'Acqua, & Jolicoeur, 2014).

The main goal of the present research was to examine the role of attention and visual working memory in the processing of simple visual displays requiring either to evaluate the number of target stimuli in the display or to verify a simple spatial relation among them. We used a MFP and EEG recordings to study the underlying mechanisms mediating performance in these tasks and in particular we relied on the SPCN to track the involvement of the visual working memory for different stimulus configurations within specific cognitive tasks.

Our hypothesis was that visual working memory would be involved to a greater degree when a given stimulus-task pairing would require more processing, compared with a stimulus-task pairing with a lighter processing demand. We expected a smaller SPCN for lighter loads. Our strategy was to combine identical stimulus configurations with two different tasks in such way that one configuration would produce a heavier load in one task and a lighter load in the other task, whereas a second configuration would produce the opposite pattern. We explain the rationale for these predictions in the remainder of the Introduction.

One task required determining how many salient visual stimuli were in a visual display. This task was based on the one used by Mazza and Caramazza (2011), which we called 'Count' (C) in this article. Using displays like the one in Fig. 1, we included either one coloured item (either green or blue) or two coloured items (two green, two blue, or one green and one blue). In the context of the MFP, subjects performed one block of trials in which a target was defined as a display containing a single coloured item, regardless of colour. In a second block of trials, a target was defined as a display containing two coloured items, regardless of colour. Subjects reported how many targets (displays matching the target definition) were seen after viewing a set of six displays. We expected to replicate the findings of Mazza and Caramazza (2011), concerning the difference between displays containing one versus two items, namely a larger N2pc and SPCN for displays with two items, regardless of target definition (one vs. two). More importantly for our present purposes, however, was that we predicted that determining that a display contained two same-colour salient items would require more time, and hence engage attention and visual working memory to a greater extent, than for two different-colour items. We base this prediction on the notion that target individuation should be facilitated by the presence of a clear perceptual cue indicating the presence of two targets (namely, two different colours). The presence of a single salient colour could be consistent with the presence of a single item or the presence of two same-colour targets. Therefore, we predicted that individuating two same-colour targets would take longer and likely require more attention and perhaps rely on visual working memory to a greater extent than the individuation of two items in distinct salient colours.

The second task, called 'Spatial Relations' (SR) required the verification of a particular spatial relation among items that had distinct colours. Here a target was defined either as a green item above a blue one, in one block of trials; or as a blue item above a green one, in another block of trials. In this task, two blue or two green items were non-targets in the MFP. In this task, we predicted that more processing, and hence attention and involvement of visual working memory, would be required when the display contained two different-colour items than for two same-colour items. If the display did not contain two colours, then it could not be a target. If, however, there were two salient colours, then the display would need to be checked for the particular spatial relation in the target definition (e.g., blue above green).

The same stimuli (pairs of same-colour or different-colour items) should thus engage attention and visual working memory to different degrees, depending on the task. This should be reflected in a larger N2pc for same-colour pairs than different-colour pairs for the Count condition while the opposite pattern should be found in the Spatial Relations condition. Similarly, the SPCN should be larger for different-colour pairs than same-colour pairs for the Spatial Relations condition while the opposite should be found in the Count condition.

2. Methodology

2.1. Subjects

Thirty-five students from Université de Montréal participated in our study; each received $20 (CAD) as compensation for their time. Data from one subject was removed from analysis due to a high number of movement artefacts. The final group consisted of 34 students (10 males, 30 right-handed, age between 18 and 34 years-old, mean = 23.59 ± 4.0 years). All subjects reported no history of neurological, psychiatric, or toxicological problems. All had normal colour vision, based on testing with diagnostic chromatic plates, and gave written informed consent prior to testing.

2.2. Stimuli and procedure

Each frame consisted of 16 items, each shaped like a digital number 8 ($0.8^\circ \times 0.4^\circ$) placed on an imaginary circle (diameter = 7.6°). The circle was centered on a fixation cross at the center of the screen, as illustrated in Fig. 1. Coloured items were blue or green while distracters were grey, all on a black background. The luminance of the blue, green, and grey figure-eight characters was equated using a Minolta CS-100 chromameter. Each trial used a MFP consisting of 6 frames. There were 3 possible types of frames: 1 item of colour (either blue or green), 2 items of the same colour (2 blue or 2 green), or 2 items of different colours (1 blue and 1 green). In each task, there were 0, 2, 4, or 6 targets in a sequence of 6 frames. When a frame contained two coloured items, they were always separated by two grey items.

Subjects were seated in a dimly lit, electrically shielded room and were positioned on a chin rest 57 cm from the computer monitor. Each trial was initiated by pressing the space bar. A white fixation cross was presented in the center of the screen for 500 ms followed by the 6 frames presented for 200 ms each. The stimulus onset asynchrony (SOA) from one frame to the next was 800 ± 100 ms. After the last frame, subjects had 2000 ms to answer before a feedback display was provided for 750 ms. Response was given by pressing a key (c, v, b, n) to indicate the number of targets seen across the 6 preceding frames on that trial (c = 0, v = 2, b = 4, or n = 6; these keys were adjacent and arranged in that order from left to right, on the bottom row of the keyboard). The experiment consisted of 4 blocks, each with a different target definition, divided into two conditions: the C condition (during which subjects had to count the number of coloured items in each frame, which could be either 1 or 2), and the SR condition (during which subjects had to identify the spatial relation between the coloured item on each frame). The general instructions were the same for each block: subjects had to count the number of frames that included a target. What defined a target changed for each block and was described in the written instructions at the beginning of each block. Possible targets for the C condition were 1 coloured item (regardless of colour) or 2 coloured items (regardless of whether the 2 items were the same colour or not, and without regard to spatial position). Possible targets for the SR condition were one blue item above a green item or one green item above a blue item (as illustrated in Fig. 1). All frames in the SR condition contained 2 items. Each block consisted of 16 practice trials (which were not included in the analysis) and 64 experimental trials for a total of 256 trials, yielding measurements for 1536 frames. The order of trials within blocks was random and the order of blocks and the hand of response were counterbalanced across subjects.

2.3. EEG recording and analysis

EEG data were recorded with 64 Ag-AgCl active electrodes mounted on an elastic cap (BioSemi Active Two system) according to the 10–10 international system (Sharbrough et al., 1991) at Fp1, Fpz, Fp2, AF7, AF3, AFz, AF4, AF8, F7, F5, F3, F1, Fz, F2, F4, F6, F8, FT7, FC5, FC3, FC1, FCz, FC2, FC4, FC6, FT8, T7, C5, C3, C1, Cz, C2, C4, C6, T8, TP7, CP5, CP3, CP1, CPz, CP2, CP4, CP6, TP8, P9, P7, P5, P3, P1, Pz, P2, P4, P6, P8, P10, PO7, PO3, POz, PO4, PO8, O1, Oz, O2, and Iz sites. Two additional electrodes, one at the left and one at the right mastoid were used, and potentials at other electrodes were re-referenced to their average. Eye movements were measured with horizontal (HEOG) and vertical electrooculogram (VEOG). HEOG was defined as the voltage difference between 2 electrodes placed at the external canthi of the eyes while VEOG was defined as the voltage difference between the signal at Fp1 and at an electrode placed below the left eye. Signals were digitized at 512 Hz (DC to 134 Hz) and later bandpass filtered from 0.05 to 30 Hz during post-recording processing. One subject had different filtering parameters (bandpass of 0.2–20 Hz) because of larger slow drifts and other noise sources compared with other subjects. Frames with incorrect answers, eye movements (a variation of 35 μV in a 300 ms interval in the HEOG difference wave), blinks (a variation of 50 μV in a 150 ms interval in the VEOG difference wave), and other artefacts were excluded from the analysis. All variations of ±100 ms in the segmentation interval were considered as an artefact. If artefacts were found in 7 or less electrodes, signals from surrounding electrodes were interpolated using the spherical spline interpolation method. If 8 or more electrodes showed an artefact, the trial was removed from analysis. The electrodes Fp1, Fpz, and Fp2 were excluded of the interpolation pool since the signal in that area is typically noisier. The EEG was segmented into 700-ms epochs starting 100 ms before and ending 600 ms after the onset of each frame.

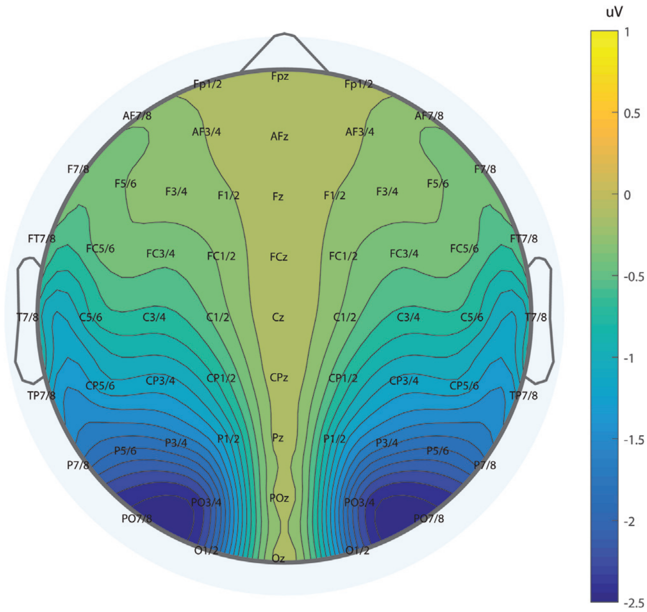

Fig. 2. Topographical map of the N2pc, top view, between 200 and 260 ms, all conditions combined.

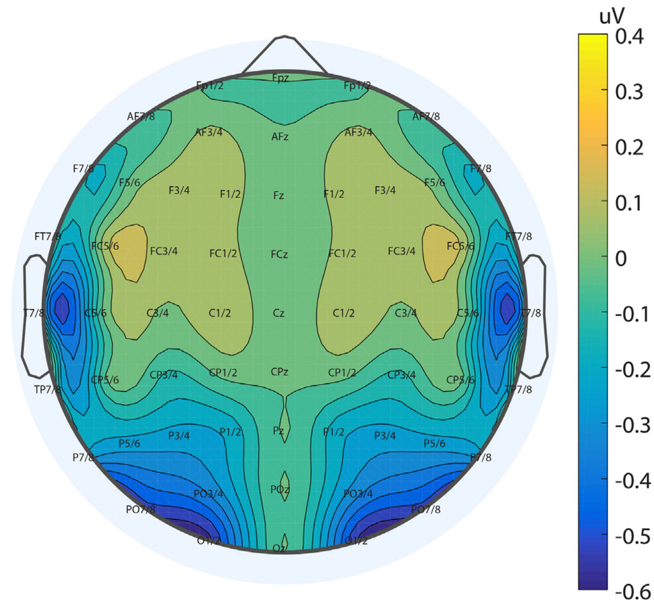

Fig. 3. Topographical map of the SPCN, top view, between 450 and 550 ms, all conditions combined.

A baseline correction was performed by subtracting the mean voltage during the 100-ms pre-frame interval from the voltage on the whole segment. Event-related lateralizations were computed by subtracting ipsilateral activity from contralateral activity for each pair of lateral electrodes. The N2pc and SPCN components were examined primarily at posterior electrodes (PO7/PO8, PO3/PO4, P7/P8, O1/O2). We report detailed results for PO7/PO8 where the N2pc and SPCN were at or near maximum.

3. Results

Only trials with correct answers were included in the analyses. Mean accuracy for the group was 96.2%. We note that a correct response summarized performance for 6 frames. Obtaining a correct response entailed making 6 consecutive correct internal decisions, or 4 correct decisions and two incorrect ones that cancelled out (miscounting a non-target as a target compensated by missing a target), with other possibilities having negligible probabilities. Using a simple binomial model, we estimated the probability of a correct decision at a frame-by-frame level at $p > 0.99$ for the present level of accuracy (the proof is left as an exercise for the reader). The mean accuracy in the C condition was 95.9% in the C condition, and 96.7% in the SR condition, $F(1, 32) = 1.83$, $p > 0.19$, showing that the tasks were not different in term of accuracy.

Our analyses of ERPs focused on lateralized potentials, specifically the N2pc and SPCN components. Figs. 2 and 3 show the topographical maps of the N2pc and SPCN respectively. Fig. 4 shows the overall event-related lateralization waveforms for each individual frames, averaging across all conditions. We first examined how the results varied across frame positions in the MFP sequence. ANOVAs were performed on the mean amplitude of the N2pc (in the time interval of 215–255 ms) to check for possible differences between the frames. We performed a first analysis for the C condition that compared results for one item versus for two items. Results for the Number of stimuli × Frames ANOVA show no significant interaction ($F(5, 165) = 1.26$, $p > 0.28$) nor main effect for the frame ($F(5, 165) = 1.17$, $p > 0.32$) for N2pc amplitude. Next we compared the C and SR conditions, for displays with two salient items. The ANOVA had Conditions × Type of stimuli (same colour vs. different colour) × Frame as within-subjects factors. There were no significant interactions involving Frame: all Fs < 1.83, all ps > 0.10.

We also examined the onset latency of the N2pc across conditions and frames using a jackknife approach in which we measured the latency at which N2pc reached 50% of the area under the curve (Kiesel et al., 2008 Kiesel, Miller, Jolicœur, & Brisson, 2008). There was only one case in which we found a significant effect, which is shown in Fig. 2: the onset of the N2pc for the first frame position was significantly delayed relative to that of the other frames, ($F(1, 33) = 12.74$, $p < 0.0012$) but only the first frame differed from the others. This was the only significant effect of Frame, and given that this effect did not interact with anything else, we did not consider frame position further. The analyses we report below are thus based on analyses in which we aggregated the data across frame positions.

Fig. 5 shows the grand average waveforms for each type of target for the C condition while Fig. 6 shows the grand average waveforms for the SR condition. Detailed analyses for the event-related lateralized components are described in more detail below.

3.1. N2pc

ANOVAs were performed on the mean amplitude of the N2pc, in the interval of 215–255 ms after averaging data from all the frames. The first analysis used Number of stimuli (1 vs. 2) × Type of stimulus (same vs. different colours) as factors and included only the C condition. We found a significant effect of Number of stimuli where the mean amplitude was larger when 2 items were presented compared to when there was only 1 item presented ($F(1, 33) = 16.78$, $p < 0.0003$). There was also a significant effect of the Type of stimulus in that the amplitude of the component was larger when the 2 items had the same colour compared to when they had different colours ($F(1, 33) = 12.57$, $p < 0.0012$), as shown in Fig. 5. In a separate analysis, we compared the effect of Type of stimulus while including only the SR condition. We found no difference in the amplitude of N2pc depending on whether the two items had the same or different colour ($F(1, 33) = 0.25$, $p > 0.62$), as shown in Fig. 6.

However, when we included Condition (C vs. SR) and Type of stimulus (same vs. different colour) in a combined ANOVA considering only trials with two items in the C condition, there was a significant Condition by Type of stimulus interaction, $F(1,$

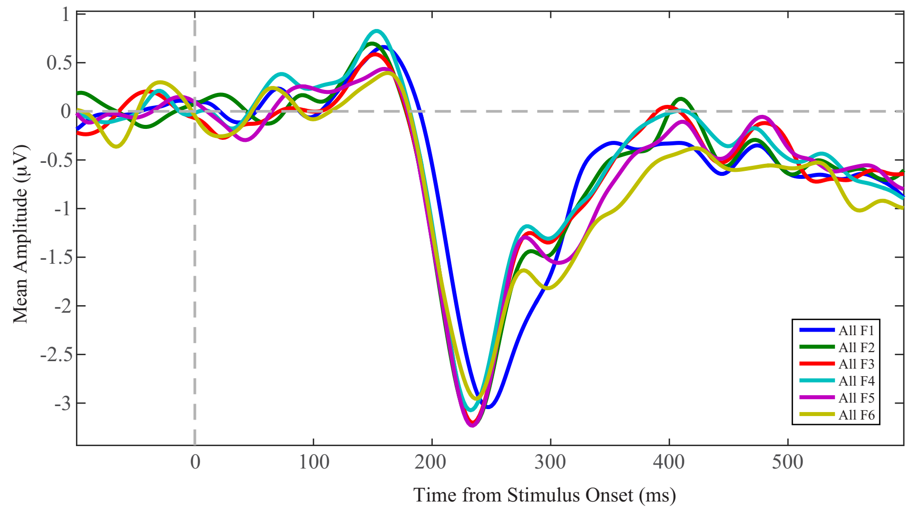

Fig. 4. Average waveforms for each frame, all conditions combined, at PO7/PO8.

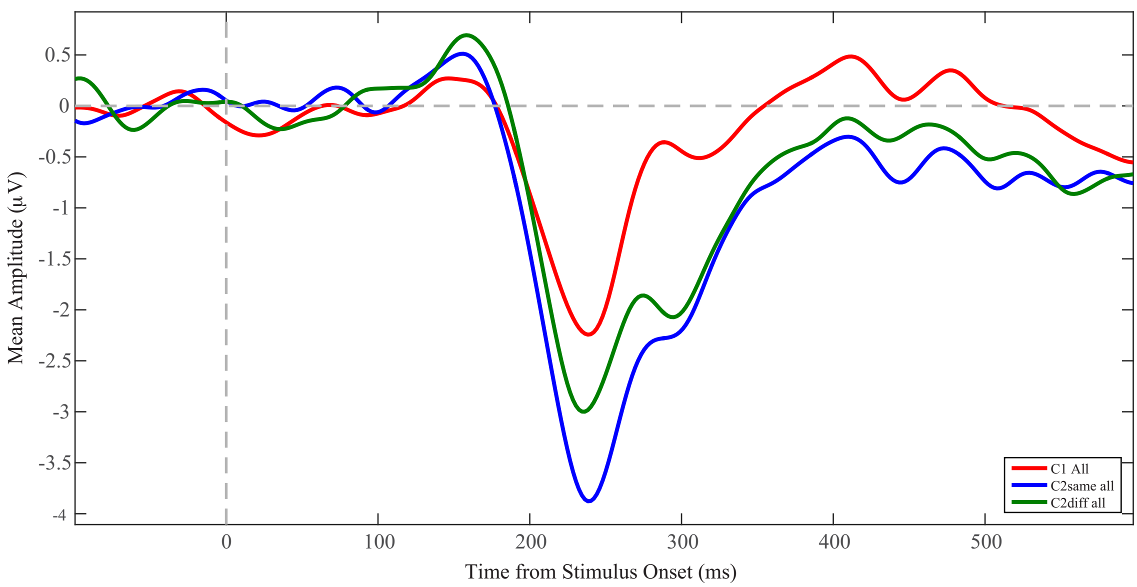

Fig. 5. Average waveforms for Count Condition at PO7/PO8.

33) = 5.53, $p < 0.025$, in which the difference between same-colour and different-colour displays was larger for the C condition than for the SR condition. We also found a main effect of the Type of stimulus ($F(1.33) = 11.93$, $p < 0.002$) where the amplitude was larger when the stimuli are the same colours compared to stimuli of different colours. There was no significant main effect of the conditions (C vs. SR) ($F(1.33) = 0.60$, $p > 0.45$).

3.2. Negative inflection

We also observed a negative inflection following the N2pc that was present for all types of stimuli. We quantified this component by taking the mean amplitude between 285 and 305 ms post-target onset. The ANOVAs presented for the N2pc component were performed using the new time window. As for N2pc, the Number of items (1 vs. 2) × Type of stimulus (same vs. different colours) analysis showed that the amplitude was larger when 2 items were presented compared to when only 1 item was presented in the C condition, $F(1, 33) = 57.43$, $p < 0.0001$. However, Figs. 5 and 6 show that it was not significantly modulated by the type of stimulus given there was no difference in this inflection when items had the same colour versus when they had different colours in both the C Condition, $F(1, 33) = 0.97$, $p > 0.33$, or the SR condition, $F(1, 33) = 3.18$, $p > 0.08$. However, the Condition (C vs. SR) × Type of stimulus (same vs. different colours) analysis showed a significant interaction $F(1, 33) = 10.95$, $p < 0.0025$, showing an inversion of the waveforms between the C and SR condition for the two different types of 2-colour trials.

3.3. SPCN

The SPCN was quantified as the mean amplitude of the event-related lateralization on the interval from 450 to 550 ms. ANOVAs were performed as for the other components. First, a Number of items × Type of Stimulus analysis showed that, in the C condition, the SPCN was larger when there were two coloured items in the frame compared to when there was only one, $F(1, 33) = 29.49$, $p < 0.0001$. There was no difference between same-colour versus different colour pairs when there were two salient items in the C condition, $F(1, 33) = 1.65$, $p > 0.20$ (see Fig. 5). However, a sig-

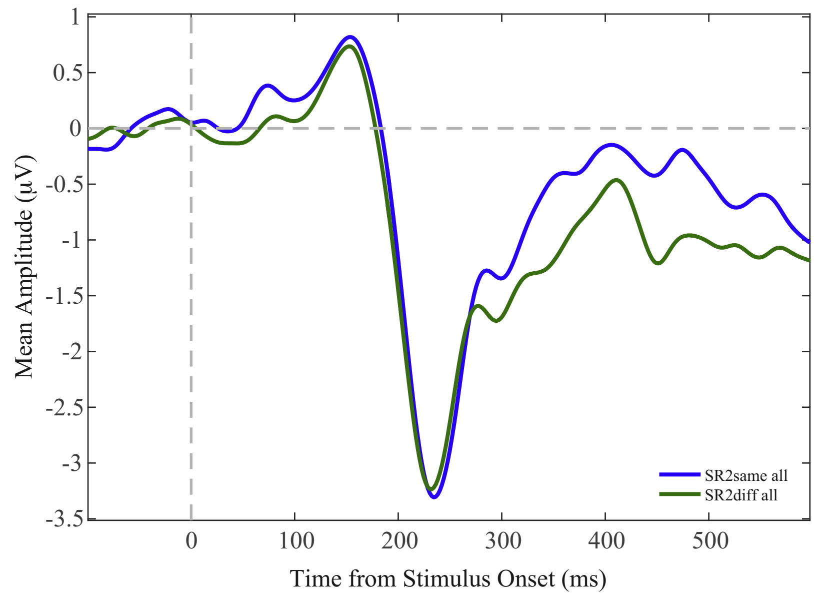

Fig. 6. Average waveforms for Spatial Relation Condition at PO7/PO8.

nificant difference was found in the SR condition in which two same-coloured items produced a smaller SPCN compared with two different-colour items, $F(1, 33) = 5.39$, $p < 0.027$ (see Fig. 6). A significant interaction Condition (C vs. SR) × Type (same-colour vs. different-colour) of stimulus was also found, $F(1, 33) = 6.38$, $p < 0.017$, reflecting the cross-over interaction in mean SPCN amplitude for different-colour and same-colour items across the SR and C conditions.

4. Discussion

The goal of this research was to examine the role of attention and visual working memory in the processing of simple visual displays requiring either counting the number of target stimuli presented or verifying a simple spatial relation among them. We used electrophysiological measures to track attention and visual working memory on a moment-to-moment basis. The stimuli were displayed using a MFP in which subjects counted the number of times particular configurations of stimuli defining a target occurred in a set of six frames. This procedure allowed us to observe electrophysiological responses to more than 1500 search displays per subject in about 60 min of time on task. The MFP, with its ability to multiply the number of trials in the study, confers an important advantage in terms of testing efficiency compared with the usual, single-frame procedure, typically used in studies of attention and memory.

We studied the cognitive processes linked to processing two different types of targets: looking for a particular number of items or identifying a specific spatial relation between two stimuli. In trial blocks in which targets were defined by a particular number of coloured items, the amplitude of the N2pc and of the SPCN were larger for two items than for one. These results replicate and extend previous findings by showing that determining that a display contains two same-colour items requires a stronger attentional response than for two different-colour items (e.g., Mazza & Caramazza, 2011, 2012). When the task was to determine whether there was a single or two coloured items, a task similar to the one used by Mazza and Caramazza, (2011, 2012), the presence of two distinct colours (green and blue) in the visual field may have allowed subjects to disengage spatial attention more quickly than when the two coloured items had the same colour. Determining if 1 or 2 items were present would be easier if the items were in different colours because there could be a maximum of 2 items and colours (i.e., if the participant noticed two different colours, they could infer that there were two items without knowing where they were). In contrast, if they only saw one colour, there could be one or two items, and a more detailed individuation process would be required to distinguish between 2 items of the same colour. We hypothesize that this greater need for attentional individuation explains the difference in the amplitude of the N2pc in the Count condition. An alternative way to think about these findings might be that individuating two different-colour items is easier than for same-colour items because of the convergence of two sources of information (e.g., discontinuities in a spatial saliency map, and discontinuities in different portions of colour space). Two locations plus two colours would create a redundancy gain relative to two locations in only one colour, facilitating individuation and reducing the need for spatial attention.

An important goal of the present work was to examine how attention and memory participate in the SR task. There were always 2 items in this condition, and targets were defined as an item of one colour above one of the other colour. We supposed that attention would be deployed on the items long enough to determine if the items had the same colour or not. If the colours were different, then we supposed that they would process the information further to determine if the correct spatial relation was present. We hypothesized that some of this processing would occur after the N2pc and require passage into visual working memory, which would be reflected by a larger amplitude of the SPCN. Importantly, this is exactly what was observed: the SPCN was larger for two items with a different colour, than for items that had the same colour. Interestingly, the pattern of SPCN amplitudes was reversed in the C condition, which required counting but not establishing a spatial relation.

We chose to use the same stimuli in two different contexts to be able to compare directly their impacts on both ERPs. The significant interaction between Condition (C vs. SR) and Type of stimulus (same vs. different-colour) in the amplitude of the SPCN demonstrates the importance of the task in determining the role of processing in visual working memory for identical stimulus configurations. We believe that these differences observed in the context of our experiment, that is, the different amplitude modulations on the N2pc and the SPCN, reflect strategic processing differences across tasks. Mazza and Caramazza (2012) used a task similar to our Count condition in which participants had to report the number of targets presented in a display composed of coloured diamonds. In their experiment, there were one, two, or three targets. Targets were defined as a diamonds in a specific colour, and there were four colours. The targets could all have unique colours, or the same colour, in different trials. The authors found a modulation on the N2pc depending on the number of targets, namely an increase in the amplitude of N2pc as the number items was increased, which we replicated in our study. However, they saw no modulation on the SPCN component depending on whether the items all had the same colour or all different colours. One may wonder why we found a larger SPCN in the Count conditions when two items had the same colour, whereas Mazza and Caramazza did not. Neither did they find a difference in SPCN between same and different colours when there were three items in the display. We suspect that their task and stimulus conditions made the simple strategy we hypothesized for our experiment less effective. Whereas the presence of different colours in our displays signalled a unique answer ('two'), different colours in the Mazza-Caramazza design could be consistent with two or three items in the display. Thus, perceiving more than one colour was not sufficient to solve the problem posed by the task-stimulus contingencies in their study. We suspect that, in their case, the most expedient strategy would be to pass information into visual working memory for further processing in all cases, whether the colours were the same or different. Notwithstanding these differences, we believe that the results show the importance of the context (or task) on how the visual and spatial information are processed in the brain.

An important finding in our study was that we found the same stimulus-task interaction during the N2pc time window. This interaction suggests that perceptuo-cognitive processing choices needed to perform the task were already taking place and were modulating the N2pc. The differences observed are unlikely to have been caused by physical properties of the stimuli because the same stimuli were used in the both conditions but did not cause the same amplitude difference on both the N2pc and the SPCN, supporting our hypothesis that the differences are caused by a different processing of the stimuli.

An unexpected result was the negative inflection found between the N2pc and the SPCN (at about 300 ms). What this inflection reflects is unclear and is not always seen in published N2pc-SPCN waveforms. However, visual inspection of Mazza and Caramazza's (2011, 2012) results shows a similar inflection, but the authors did not discuss it. The component was present in all conditions and types of stimuli in the present article. We initially thought that this inflection might be a second and smaller N2pc linked to the shift of attention to the second item presented. However the presence of the inflection when there was only 1 item in the C condition makes this interpretation rather implausible. We leave it to future research to determine if there is a functional significance associated with this deflection.

Finally, the results highlight the success of the MFP. The main advantage of the procedure is that it typically increases the number of studied attentional events by a factor of two or more compared with a typical single-frame procedure. This increased efficiency can be put to good use in the laboratory by enabling more sensitive designs or designs with more conditions. In the clinic, the MFP holds the promise of enabling shorter tests without loss of sensitivity, which is particularly important for patients with limited stamina, tight schedules, and/or use with expensive technologies.

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